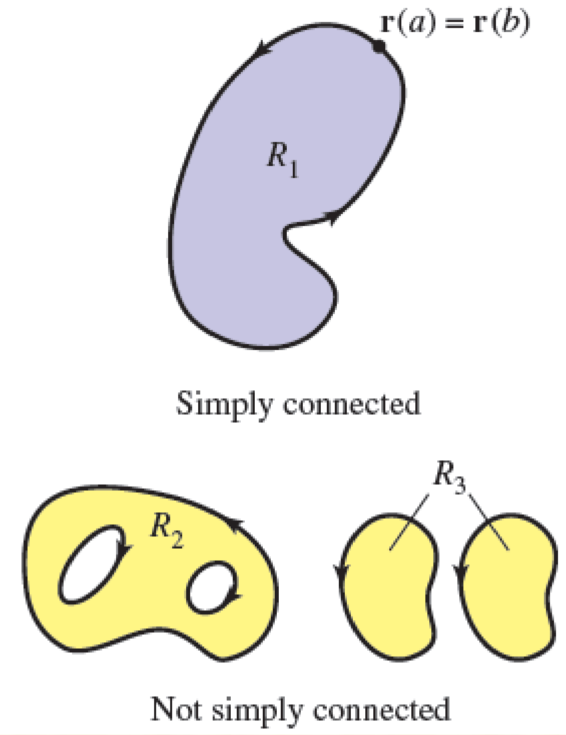
Green's Theorem

A connected plane region R is **simply connected** when every simple closed curve in R encloses only points that are in R (see Figure 15.26).

Informally, a simply connected region cannot consist of separate parts or holes.



Green's Theorem

DEFINITION **Divergence (Flux Density)**

The **divergence (flux density)** of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ at the point (x, y) is

$$\operatorname{div} \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}. \tag{1}$$

If F denotes the velocity of a fluid in a medium. Then if div(F)=0, fluid is said to be incompressible.

In electromagnetic theory, if div(F)=0, then vector field F is said to be solenoidal (magnetic).

Green's Theorem

DEFINITION **k-Component of Curl (Circulation Density)**

The k-component of the curl (circulation density) of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ at the point (x, y) is the scalar

$$(\text{curl }\mathbf{F}) \cdot \mathbf{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}. \tag{2}$$

THEOREM 3 **Green's Theorem (Flux-Divergence or Normal Form)**

The outward flux of a field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ across a simple closed curve C equals the double integral of div $\mathbf{F}$ over the region R enclosed by C.

$$\underbrace{\oint_C \mathbf{F} \cdot \mathbf{n}\, ds}_{\text{Outward flux}} = \oint_C M\, dy - N\, dx = \underbrace{\iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}\right) dx\, dy}_{\text{Divergence integral}} \tag{3}$$

Green's Theorem

THEOREM 4 Green's Theorem (Circulation-Curl or Tangential Form)

The counterclockwise circulation of a field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ around a simple closed curve C in the plane equals the double integral of $(\text{curl } \mathbf{F}) \cdot \mathbf{k}$ over the region R enclosed by C.

$$\underbrace{\oint_C \mathbf{F} \cdot \mathbf{T}\, ds = \oint_C M\, dx + N\, dy}_{\text{Counterclockwise circulation}} = \underbrace{\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx\, dy}_{\text{Curl integral}} \tag{4}$$

Green's Theorem

EXAMPLE 3 Supporting Green's Theorem

Verify both forms of Green's Theorem for the field

$$\mathbf{F}(x, y) = (x - y)\mathbf{i} + x\mathbf{j}$$

and the region R bounded by the unit circle

$$C: \quad \mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}, \qquad 0 \leq t \leq 2\pi.$$

Solution We have

$$M = \cos t - \sin t, \qquad dx = d(\cos t) = -\sin t\, dt,$$

$$N = \cos t, \qquad dy = d(\sin t) = \cos t\, dt,$$

$$\frac{\partial M}{\partial x} = 1, \qquad \frac{\partial M}{\partial y} = -1, \qquad \frac{\partial N}{\partial x} = 1, \qquad \frac{\partial N}{\partial y} = 0.$$

Green's Theorem

The two sides of Equation (3) are

$$\oint_C M\,dy - N\,dx = \int_{t=0}^{t=2\pi} (\cos t - \sin t)(\cos t\,dt) - (\cos t)(-\sin t\,dt)$$

$$= \int_0^{2\pi} \cos^2 t\,dt = \pi$$

$$\iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}\right) dx\,dy = \iint_R (1 + 0)\,dx\,dy$$

$$= \iint_R dx\,dy = \text{area inside the unit circle} = \pi.$$

The two sides of Equation (4) are

$$\oint_C M\,dx + N\,dy = \int_{t=0}^{t=2\pi} (\cos t - \sin t)(-\sin t\,dt) + (\cos t)(\cos t\,dt)$$

$$= \int_0^{2\pi} (-\sin t \cos t + 1)\,dt = 2\pi$$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{dy}\right) dx\,dy = \iint_R (1 - (-1))\,dx\,dy = 2\iint_R dx\,dy = 2\pi.$$

Green's Theorem

Example 5: Using Green's theorem calculates the outward flux of the vector field $F = xi + y^2 j$ across the square bounded by the lines $x = \pm 1$ and $y = \pm 1$.

Solution Calculating the flux with a line integral would take four integrations, one for each side of the square. With Green's Theorem, we can change the line integral to one double integral. With $M = x$, $N = y^2$, C the square, and R the square's interior, we have

$$\begin{aligned}
\text{Flux} &= \oint_C \mathbf{F} \cdot \mathbf{n}\, ds = \oint_C M\, dy - N\, dx \\
&= \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx\, dy \qquad \text{Green's Theorem} \\
&= \int_{-1}^{1} \int_{-1}^{1} (1 + 2y)\, dx\, dy = \int_{-1}^{1} \Big[x + 2xy \Big]_{x=-1}^{x=1} dy \\
&= \int_{-1}^{1} (2 + 4y)\, dy = \Big[2y + 2y^2 \Big]_{-1}^{1} = 4.
\end{aligned}$$

Divergence Theorem

THEOREM 15.12 The Divergence Theorem

Let Q be a solid region bounded by a closed surface S oriented by a unit normal vector directed outward from Q. If $\mathbf{F}$ is a vector field whose component functions have continuous first partial derivatives in Q, then

$$\int\!\!\int_S \mathbf{F} \cdot \mathbf{N}\, dS = \int\!\!\int\!\!\int_Q \operatorname{div} \mathbf{F}\, dV.$$

Divergence Theorem

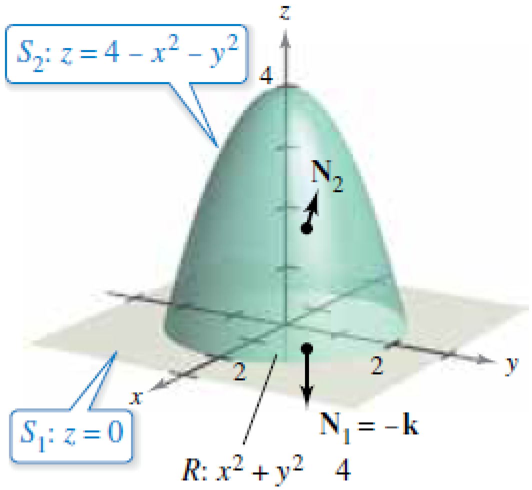


Figure 15.58

EXAMPLE 2 **Verifying the Divergence Theorem**

Let Q be the solid region between the paraboloid

$$z = 4 - x^2 - y^2$$

and the xy-plane. Verify the Divergence Theorem for

$$\mathbf{F}(x, y, z) = 2z\mathbf{i} + x\mathbf{j} + y^2\mathbf{k}.$$

Solution From Figure 15.58, you can see that the outward normal vector for the surface S_1 is $\mathbf{N}_1 = -\mathbf{k}$, whereas the outward normal vector for the surface S_2 is

$$\mathbf{N}_2 = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}.$$

So, by Theorem 15.11, you have

$$\begin{aligned}
\int_S\!\!\int \mathbf{F} \cdot \mathbf{N}\, dS &= \int_{S_1}\!\!\int \mathbf{F} \cdot \mathbf{N}_1\, dS + \int_{S_2}\!\!\int \mathbf{F} \cdot \mathbf{N}_2\, dS \\
&= \int_{S_1}\!\!\int \mathbf{F} \cdot (-\mathbf{k})\, dS + \int_{S_2}\!\!\int \mathbf{F} \cdot \frac{(2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k})}{\sqrt{4x^2 + 4y^2 + 1}}\, dS \\
&= \int_R\!\!\int -y^2\, dA + \int_R\!\!\int (4xz + 2xy + y^2)\, dA \\
&= -\int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} y^2\, dx\, dy + \int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} (4xz + 2xy + y^2)\, dx\, dy
\end{aligned}$$

Divergence Theorem

$$
\begin{aligned}
&= \int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} (4xz + 2xy)\, dx\, dy \\
&= \int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} [4x(4 - x^2 - y^2) + 2xy]\, dx\, dy \\
&= \int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} (16x - 4x^3 - 4xy^2 + 2xy)\, dx\, dy \\
&= \int_{-2}^{2}\Big[8x^2 - x^4 - 2x^2y^2 + x^2y\Big]_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} dy \\
&= \int_{-2}^{2} 0\, dy \\
&= 0.
\end{aligned}
$$

On the other hand, because

$$
\begin{aligned}
\text{div } \mathbf{F} &= \frac{\partial}{\partial x}[2z] + \frac{\partial}{\partial y}[x] + \frac{\partial}{\partial z}[y^2] \\
&= 0 + 0 + 0 \\
&= 0
\end{aligned}
$$

you can apply the Divergence Theorem to obtain the equivalent result

$$
\begin{aligned}
\int_S\!\!\int \mathbf{F} \cdot \mathbf{N}\, dS &= \iiint\limits_Q \text{div } \mathbf{F}\, dV \\
&= \iiint\limits_Q 0\, dV \\
&= 0.
\end{aligned}
$$

Example 6 – Using the Divergence Theorem

Let Q be the solid region bounded by the coordinate planes and the plane $2x + 2y + z = 6$ and let $\mathbf{F} = x\mathbf{i} + y^2\mathbf{j} + z\mathbf{k}$.

Find $\displaystyle\int_S\int \mathbf{F} \cdot \mathbf{N}\, dS$ where S is the surface of Q.

Solution: From Figure 15.57, you can see that Q is bounded by four subsurfaces.

So, you would need four surface integrals to evaluate

$$\int_S\int \mathbf{F} \cdot \mathbf{N}\, dS.$$

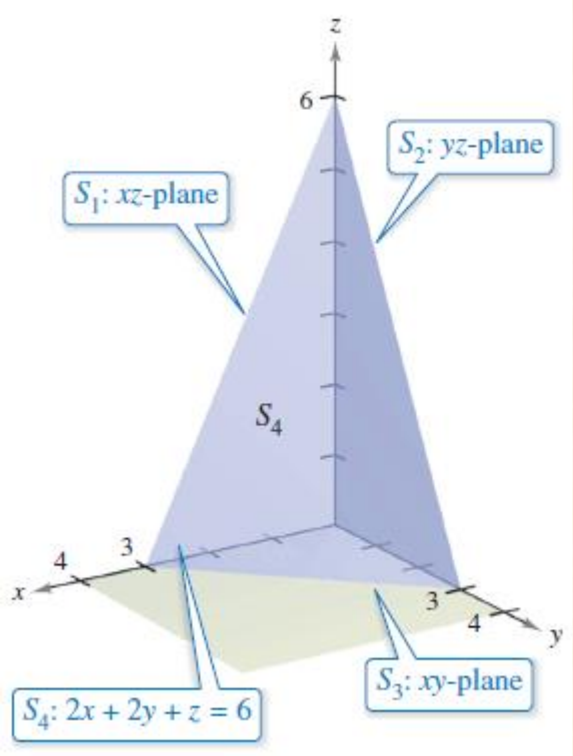


Figure 15.57

Example 6 – Using the Divergence Theorem

Solution: However, by the Divergence Theorem, you need only one triple integral. Because

$$\operatorname{div}\mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}$$

$$= 1 + 2y + 1 \quad = 2 + 2y$$

you have

$$\int\!\!\int_S \mathbf{F}\cdot\mathbf{N}\,dS = \int\!\!\int\!\!\int_Q \operatorname{div}\mathbf{F}\,dV$$

$$= \int_0^3 \int_0^{3-y} \int_0^{6-2x-2y} (2 + 2y)\,dz\,dx\,dy$$

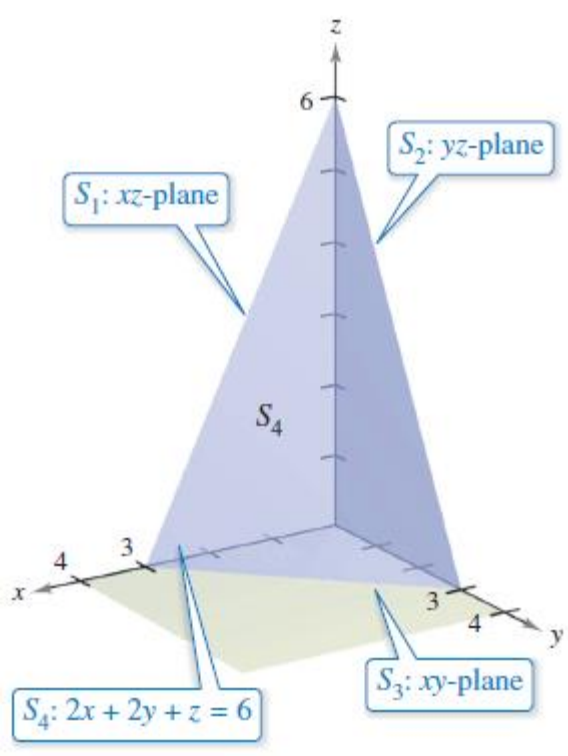


Figure 15.57

Example 6 – Using the Divergence Theorem

$$= \int_0^3 \int_0^{3-y} (2z + 2yz)\Big]_0^{6-2x-2y} dx\,dy$$

$$= \int_0^3 \int_0^{3-y} (12 - 4x + 8y - 4xy - 4y^2)\,dx\,dy$$

$$= \int_0^3 \left[12x - 2x^2 + 8xy - 2x^2y - 4xy^2\right]_0^{3-y} dy$$

$$= \int_0^3 (18 + 6y - 10y^2 + 2y^3)\,dy$$

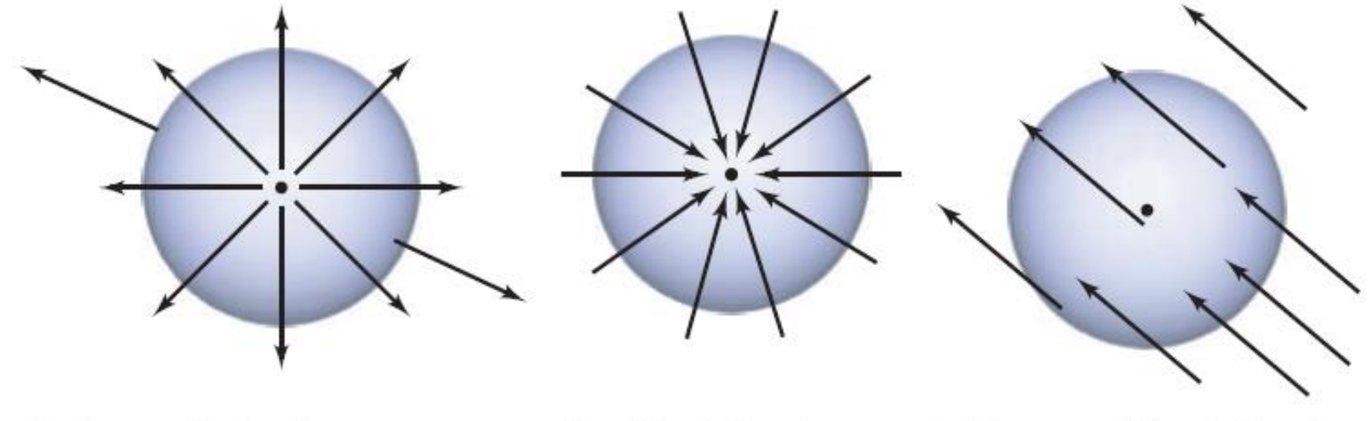
$$= \left[18y + 3y^2 - \frac{10y^3}{3} + \frac{y^4}{2}\right]_0^3$$

$$= \frac{63}{2}.$$

Flux and the Divergence Theorem

The point (x_0, y_0, z_0) in a vector field is classified as a source, a sink, or incompressible, as shown in the list below.

1. **Source,** for div **F** > 0
2. **Sink,** for div **F** < 0
3. **Incompressible,** for div **F** = 0



(a) Source: div **F** > 0 (b) Sink: div **F** < 0 (c) Incompressible: div **F** = 0

Figure 15.61

Example 7 – Calculating Flux by the Divergence Theorem

Let Q be the region bounded by the sphere $x^2 + y^2 + z^2 = 4$. Find the outward flux of the vector field $\mathbf{F}(x, y, z) = 2x^3\mathbf{i} + 2y^3\mathbf{j} + 2z^3\mathbf{k}$ through the sphere.

Solution:

By the Divergence Theorem, you have

$$\text{Flux across } S = \iint_S \mathbf{F} \cdot \mathbf{N}\, dS \quad = \iiint_Q \operatorname{div} \mathbf{F}\, dV \quad = \iiint_Q 6(x^2 + y^2 + z^2)\, dV.$$

Next, use spherical coordinates with $\rho^2 = x^2 + y^2 + z^2$ and $dV = \rho^2 \sin\phi\, d\theta\, d\phi\, d\rho.$

$$\iiint_Q 6(x^2 + y^2 + z^2)\, dV \quad = 6\int_0^2 \int_0^{\pi} \int_0^{2\pi} \rho^4 \sin\phi\, d\theta\, d\phi\, d\rho$$

Spherical coordinates

$$= 6\int_0^2 \int_0^{\pi} 2\pi\rho^4 \sin\phi\, d\phi\, d\rho \quad = 12\pi \int_0^2 2\rho^4\, d\rho \quad = 24\pi\left(\frac{32}{5}\right) \quad = \frac{768\pi}{5}.$$

Stokes' Theorem

THEOREM 5 Stokes' Theorem

The circulation of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ around the boundary C of an oriented surface S in the direction counterclockwise with respect to the surface's unit normal vector $\mathbf{n}$ equals the integral of $\nabla \times \mathbf{F} \cdot \mathbf{n}$ over S.

$$\underbrace{\oint_C \mathbf{F} \cdot d\mathbf{r}}_{\text{Counterclockwise circulation}} = \underbrace{\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n}\, d\sigma}_{\text{Curl integral}} \tag{4}$$

Stokes' Theorem

EXAMPLE 6: A fluid of constant density rotates around the z-axis with velocity F = $w(-yi + xj)$, where w is a positive constant called the *angular velocity* of the rotation (Figure 16.61). Find ∇ X F and relate it to the circulation density.

Solution With $\mathbf{F} = -\omega y\mathbf{i} + \omega x\mathbf{j}$, we find the curl

$$\nabla \times \mathbf{F} = \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z}\right)\mathbf{i} + \left(\frac{\partial M}{\partial z} - \frac{\partial P}{\partial x}\right)\mathbf{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\mathbf{k}$$

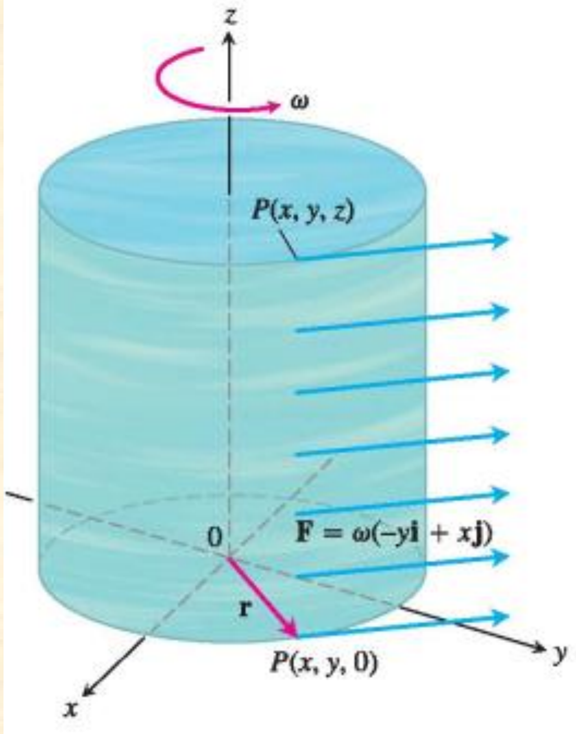
$$= (0 - 0)\mathbf{i} + (0 - 0)\mathbf{j} + (\omega - (-\omega))\mathbf{k} = 2\omega\mathbf{k}.$$

By Stokes' Theorem, the circulation of $\mathbf{F}$ around a circle C of radius ρ bounding a disk S in a plane normal to $\nabla \times \mathbf{F}$, say the xy-plane, is

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \nabla \times \mathbf{F} \cdot \mathbf{n}\, d\sigma = \iint_S 2\omega\mathbf{k} \cdot \mathbf{k}\, dx\, dy = (2\omega)(\pi\rho^2).$$

Thus solving this last equation for 2ω, we have

$$(\nabla \times \mathbf{F}) \cdot \mathbf{k} = 2\omega = \frac{1}{\pi\rho^2} \oint_C \mathbf{F} \cdot d\mathbf{r},$$



Stokes' Theorem

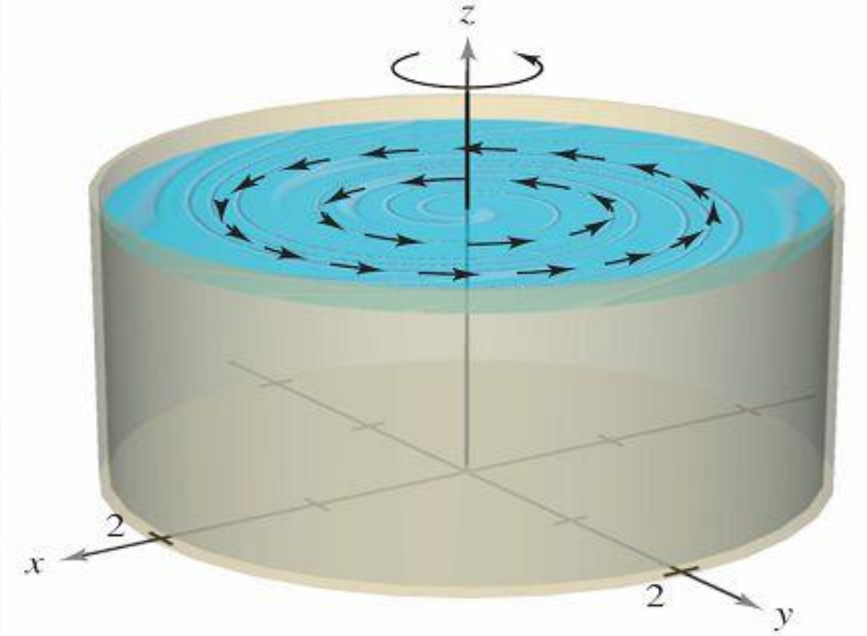
EXAMPLE 7: Use Stokes' Theorem to evaluate $\int F.dr$, if F = $xzi + xyj + 3xzk$ and C is the boundary of the portion of the plane 2x + y + z = 2 in the first octant, traversed counterclockwise as viewed from above.

Stokes' Theorem

A liquid is swirling around in a cylindrical container of radius 2, so that its motion is described by the velocity field

$$\mathbf{F}(x, y, z) = -y\sqrt{x^2 + y^2}\,\mathbf{i} + x\sqrt{x^2 + y^2}\,\mathbf{j}$$

as shown in figure at right.



Find

$$\iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{N}\, dS$$

where S is the upper surface of the cylindrical container.

Stokes' Theorem

The curl of **F** is given by

$$\text{curl}\,\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ -y\sqrt{x^2+y^2} & x\sqrt{x^2+y^2} & 0 \end{vmatrix}$$

$$= 3\sqrt{x^2+y^2}\,\mathbf{k}.$$

Letting **N** = **k**, you have

$$\int_S\int (\text{curl}\,\mathbf{F}) \cdot \mathbf{N}\, dS = \int_R\int 3\sqrt{x^2+y^2}\, dA$$

Stokes' Theorem

$$= \int_0^{2\pi} \int_0^2 (3r)r \, dr \, d\theta$$

$$= \int_0^{2\pi} r^3 \Big]_0^2 \, d\theta$$

$$= \int_0^{2\pi} 8 \, d\theta$$

$$= 16\pi.$$

SUMMARY

SUMMARY OF INTEGRATION FORMULAS

Fundamental Theorem of Calculus

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$

Fundamental Theorem of Line Integrals

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(x(b), y(b)) - f(x(a), y(a))$$

Green's Theorem

$$\int_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA = \int_C \mathbf{F} \cdot \mathbf{T}\,ds = \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_R (\text{curl}\,\mathbf{F}) \cdot \mathbf{k}\,dA$$

$$\int_C \mathbf{F} \cdot \mathbf{N}\,ds = \iint_R \text{div}\,\mathbf{F}\,dA$$

Divergence Theorem

$$\iint_S \mathbf{F} \cdot \mathbf{N}\,dS = \iiint_Q \text{div}\,\mathbf{F}\,dV$$

Stokes's Theorem

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl}\,\mathbf{F}) \cdot \mathbf{N}\,dS$$

Thanks a lot …